

Live-Colour Detection Using MATLAB

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Many real-world applications like motion detection, object detection and colour detection require real-time image processing. Here, we present a method to detect live-coloured images or objects using a camera and MATLAB program. If any RGB (red, green and blue) colour comes in front of the camera, it detects a particular colour depending on the user's command. This concept can be used in many

applications like colour-object finder and colour-detector robot, among others.

Image processing

Image processing is a method of converting an image into digital form and performing operations on it to get an enhanced image or to extract some useful information from it. Most image-processing techniques involve treating the image as a two-dimensional signal and applying standard signal-processing techniques to it.

Colour-detection algorithm

In the colour-detection method, primary-colour (red, green, blue) objects can be detected easily. The algorithm used for colour detection actually works in separate steps as given below:

1. Take a snapshot of the original

red object/image.

2. Convert that original snapshot to grey image.

3. Extract red component from the original snapshot.

4. Subtract red-colour components from the grey image. (We will get another grey image.)

5. Remove noise from the new grey image by using filter command.

6. Convert the filtered image into binary image using binary command. (We will get a bright image in place of the red object.)

7. Measure parameters of the

bright image and place a rectangular box over it using mathematical calculations, which will show the presence of red colour at that place.

Similarly, we detect green and blue colours by extracting green and blue colour components from the original snapshot and then subtracting those green and blue colour components from the grey-colour snapshots and follow the same from step 5 onwards.

MATLAB program

Here, the graphical user interface (GUI) is developed in MATLAB to take advantage of its features. A GUI allows users to perform tasks interactively through controls such as buttons and sliders. The main output screen of the colour-detection program is shown in Fig. 1.

The program has a main function (detection.m) among many other functions. Do not edit the functions as those are the linkers and non-executable codes. The only executable code is detection.m that uses those functions.

First, you have to find formats supported by the camera and device ID by using the following command (also shown in Fig. 2.):

```
info = imagehwinfo('winvideo')
```

This command is used to find the device ID you are using so that you can change the device ID number in your source code. In this example, our device ID is 1, so we have written 1 in the code as given below.

```
vid = videoinput('winvideo',1,'YUY2_320x240');
```

YUY2_320x240 is the resolution supported by the camera. There are other formats in MATLAB that you



Fig. 1: Main output panel of the colour-detection program

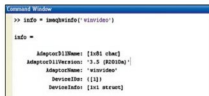


Fig. 2: Checking the device ID

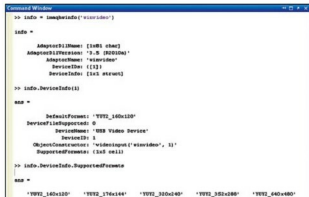


Fig. 3: Formats supported by the camera

